

Most companies, in their audio amplifier datasheets, specify the wattage that the amplifiers can provide against the speaker impedance. Designers refer to this data and select a suitable option. It is critical that amplifiers do not over-drive the speakers, as that may damage the speakers. Speakers are driven by either voltage or current. If you want to listen to mobile phone audio from the speakers of your home theatre system, it will not produce any sound because the mobile phone output (1-3W) is insufficient to drive system.

Speakers are basically inductive loads and audio is a variable AC signal. The presence of a DC signal may damage the speakers. Typically, a DC-blocking capacitor is used to restrict DC from entering the speaker.

"In the case of a faulty capacitor, you can build a DC-detection circuit on the board that will shut down the amplifier if DC is present," says Awasthy. But this does not provide any protection if the amplifier IC is damaged. Tan says, "If the amplifier inside is stilled or damaged, it will not be able to protect your speaker and your speaker might get burnt. So you must have a circuit to protect your speaker outside the amplifier."

"Choice of the method depends on the entire system design. You should follow the application guidelines provided by the component manufacturer thoroughly before making any decision," suggests Tan.

Amplifier configurations

The output transistors of the amplifier can be connected either in single-ended (SE) configuration or differential (also called bridge-tied load (BTL)) configuration. "Basically, for specific output power, for example, 15W, if you use a BTL amplifier, the supply voltage required is half that of SE. Assuming that the BTL needs 12V supply to drive 15W, for SE you may require 24V voltage supply," explains Tan.

BLT amplifiers are preferred to SE when the supply voltage is minimal.

Class D audio amplifier and support application providers

- ADI
- Maxim
- National Semiconductors
- NXP Semiconductors
- Neofidelity
- Sanyo
- STMicroelectronics
- Texas Instruments
- Wolfson
- Yamaha

Also, in SE configuration, some DC is present at the output. So with SE amplifiers, you need DC-blocking capacitors.

As mentioned earlier, the output of Class D amplifiers is an amplified PWM waveform that cannot be fed directly to the speaker. You need a second-order low-pass filter (LPF) consisting of inductor and capacitor to suppress the noise. In the case of a BLT amplifier, you need two second-order LPFs at the output against one that is sufficient for SE load configuration. Again, it totally depends on the available supply voltage and design environment.

While SE and BLT indicate the way the amplifiers and speakers are connected in an audio circuit, mono and stereophonic refer to separation of audio channels. A monophonic system consists of only one mixed audio signal that is fed to multiple speakers. The audio signal contains no information on the level and phase to replicate or simulate the directional cues. On the other hand, in a stereophonic sound, there are two independent audio channels that define a specific relationship between the level and phase at the speakers.

Modulation

PWM technique used in Class D amplifiers ideally offers better audio-band noise ratio at few hundreds of kilohertz frequency. A new technology called full-flexible (FX) amplification based on selectable PWM for configurable

products provides Hi-Fi-quality digital stream to the speakers. "Many new devices based on this technology use an all-digital approach that combines an FFX controller with a digitally-controlled power device on a single chip to both reduce cost and simplify design," shares Pagotto.

Traditionally, PWM utilises binary, or bi-state, operation. Binary operation, which consists of positive power and negative power, often wastes some power during the switching. "Damped ternary modulation consists of three switching states—positive, negative and zero power. Since it goes to zero state, it's not always in full level signals for small outputs. This saves on unwanted energy by supplying power only to produce the signal," suggests Pagotto.

Industry focus on Class D audio power amplifiers

Today, 80-90 per cent of designers have switched to Class D amplifiers for their efficiency and form factor. Many manufacturers have even discontinued some of their Class AB products. Class D amplifiers are easy to design. Also, in applications like MP3 players and mobile phones, these are preferred as they consume less battery power. Class D amplifiers are very good for low-ohmic (1-ohm and 2-ohm) loads. Product designers working on new projects are looking at Class D amplifiers instead of Class AB.

Some applications like Hi-Fi systems still use Class AB audio power amplifiers for their high fidelity, linearity and low crossover distortion. However, Class D amplifiers can provide performance comparable to Class AB with proper component selection and careful design layout. Thus, highly efficient Class D amplifiers that provide increased power density and better audio performance can be a good choice for many products and applications. ●

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